

Tanush Paradeshi

tanushparadeshi.com

linkedin.com/in/tanush-paradeshi-638754214

Email : tparades@andrew.cmu.edu

Mobile : 9493466073

github.com/tanu1239

EDUCATION

- **Carnegie Mellon University** 2024-2028
• *B.S. Computer Science* GPA: 3.83/4.0
 - **Coursework:** Data Structures and Algorithms, Computer Systems, Parallel Algorithms and Data Structures, Intro to ML, Parallel Computing and Computer Architecture, Linear Algebra, Multivariate Calculus, Probability Theory, Functional Programming, Discrete Math, Complexity Theory

EXPERIENCE

- **CMU School of Computer Science Zhang Lab** Advisor: Dr. Martin Zhang
• *Software Research Intern* Mar 2025 - Sept 2025
 - Developed a PyTorch/OpenCV pipeline for cell-count estimation from spatial transcriptomic histology image data; optimized preprocessing and evaluated predictions against expert annotations using IoU and correlation metrics
 - Developed a Bayesian inference pipeline using MCMC sampling and Metropolis-Hastings to infer cell-level gene expression from spatial transcriptomics RNA-seq data and cell-count estimates from the histology pipeline
 - Built and evaluated disease/trait association models from inferred cell-level expression; parallelized experiments using CUDA on high-performance computing (HPC) clusters to compare inference variants across tissue types
- **Stony Brook University School of Informatics IMAGINE Lab** Advisor: Dr. Prateek Prasanna
• *Summer Informatics Research Assistant* May 2024 - Sept 2024
 - Performed radiomic image analysis on dental cone-beam CT scans to analyze vascular calcifications.
 - Applied feature selection and trained/evaluated machine learning models (neural networks, random forests, logistic regression) to predict clinical outcomes such as stroke and heart attack from medical history and radiomic features
- **UC Irvine, Learning and Decision Neuroscience Lab** Advisor: Dr. Mimi Liljeholm
• *Software Research Assistant* Jun 2022 - Nov 2023
 - Designed and maintained lab's cloud-based web experiment infrastructure using Google App Engine and Firebase (Firestore + Auth), supporting large-scale online decision-making studies on Amazon MTurk and Prolific
 - Built Flask web experiments with Firebase APIs and CI/CD deployment to study neural basis of decision-making

PROJECTS

- **AnkiLite: Time-Aware Spaced Repetition Learning System** Jan 2025 - Feb 2025
 - Built a full-stack flashcard learning system that tracks user response time to estimate recall difficulty and dynamically schedules future flashcard reviews based on user accuracy and response speed.
 - Implemented a Dockerized Django REST backend with JWT authentication and an Angular frontend supporting multi-user login, card creation, deck management, and scheduling of card review order.
- **Deep Learning-Based Mobile Application for Retinitis Pigmentosa Detection** Sept 2022 - Feb 2023
 - Developed and optimized an ensemble neural network model with transfer learning (MobileNet) to diagnose retinitis pigmentosa from retinal images, achieving 97% accuracy and 0.94 F1 score on 300-image dataset.
 - Integrated the trained model into a React Native/Expo mobile app for on-device inference using TensorFlow.js
- **Localizing Optimal Locations for Deep Transcranial Magnetic Brain Stimulation** Jun 2022 - Sept 2023
 - Performed brain MRI tissue segmentation and large-scale finite element simulations of induced electric fields to identify optimal coil placement locations on scalp for maximal stimulation of target brain regions
 - Analyzed data using unsupervised machine learning (dimensionality reduction and clustering) and validated findings against an independent TMS electric field simulation toolkit developed by Massachusetts General Hospital.
 - Authored research paper: "Identifying Locations for Transcranial Magnetic Stimulation Coil Placement For Deep Brain Structures" in Journal for Student Research Vol. 12 No. 4 2023

ACHIEVEMENTS

- **USA Biology Olympiad 2023 Finalist/Top 20 Scorer (~11,000 participants nationally)**
 - Invited to training camp for top 20 scorers to compete for spot on USA International Biology Olympiad (IBO) team
- **USA National Brain Bee Neuroscience competition 5th Place (2500 participants nationally)**

SKILLS

- **Proficient Programming Languages:** Python, C/C++, Java, JavaScript, HTML/CSS, SQL, R, MATLAB
- **Tools & Frameworks:** Git, Docker, PyTorch, TensorFlow, Django, MongoDB, Flask, React Native, Angular, CUDA, scikit-learn, NumPy, Pandas, OpenCV, Google App Engine, AWS EC2, Firebase